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IMMISCIBLE ALLOYS UNDER MICROGRAVITY CONDITIONS

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ABSTRACT

The decomposition of immiscible metallic alloys into a dispersion of liquid droplets in a liquid matrix and its spatial development is considered with reference to selected experiments performed in experiment facilities under microgravity conditions. The analysis and discussion of these experiments reveal, that research in space laboratories is and will be able to clarify the mechanisms involved in the transformation of a one phase liquid into a two phase liquid. On earth many details of the different processes contributing to the decomposition are generally not observable due to the overriding effects gravity induced convections play. The various experiments so far performed in space laboratories show, that there is a need to improve our current theories for the different steps of the demixing process, to include hydrodynamic effects and droplet motions, which can have various origins, and have real importance in determining the final microstructure of the two phase alloy.

INTRODUCTION

A number of experiments have been performed under microgravity conditions with metallic alloys which are immiscible in the liquid state. They were thought of as proper candidates for experiments in space,because on earth the immiscible liquids separate into distinct layers within a few seconds due to their density difference leading to gravity induced sedimentation and convections.

The expectation that μg-conditions during experiments will lead to a spatially homogeneous dispersion of one liquid distributed in the other was, however,unfulfilled. With only a few exceptions, all metallic dispersions showed even in space massive separation. The failures had nevertheless positive consequences: they stimulated intensive theoretical and experimental research on earth and in space to clarify the mechanisms of phase separation, because the negative results of the first experiments made obvious that, even on earth, the kinetics of spatial phase separation is not solely dictated by gravity. A large number of possible additional mechanisms have been identified theoretically /1/ which may influence the spatial microstructure formation. Liquid immiscible metallic alloys have therefore gained a renewed interest for scientific and technological research. Their potential for technological applications is high.Today their use is restricted to some selected alloys,which can be processed by solidification technologies or powder metallurgical techniques. They are used for instance as electrical contact materials, power switches, welding electrodes, superconducting materials and self-lubricating bearings.

The limited success to produce such dispersions under microgravity conditions has brought about, that fast industrial applications are no longer the driving force for space research. The efforts are now concentrated on basic research concerning the whole field of liquid immiscibles to understand their behaviour during phase separation.

Therefore,the objectives of this paper are to show where experiments with the various facilities in space have extended our understanding of immiscibles, and ask new questions regarding the main points to be answered in the future.This will lead hopefully to a more unifying picture of the phase decomposition and spatial separation process. Only some selected experiments will be discussed in some detail, compared with theoretical predictions,where possible,to elucidate certain central routes of research. For a more comprehensive survey of most μg-experiments the reader should refer to the literature given in references /2/.

BASIC CONSIDERATIONS ON DECOMPOSITION IN IMMISCIBLES

Immiscible systems are by no means rare. A few hundred are known to the metallurgist. Their common feature is shown in Fig.1 illustrating a typical binary phase diagram of a monotectic alloy.
Above a critical temperature T_c both components are completely dissolvable into each other. On cooling an alloy of the gross composition c_o from a temperature $T > T_c$ the homogeneous liquid L decomposes into the liquids L_1 and L_2 when crossing the binodal line (i.e. the curve of coexistence). The composition of the two liquids at a temperature $T_m < T_o < T_c$ is given by the phase diagram and the volume fraction of the minority phase L_2 and the matrix phase L_1 are given by the lever rule. On

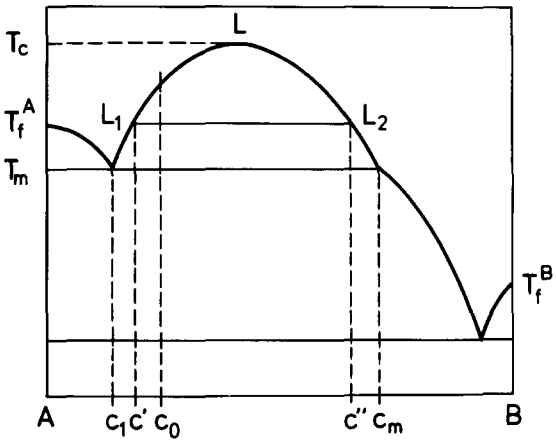

Fig.1 Schematic binary phase diagram of an alloy exhibiting a miscibility gap in the liquid state. Alloys with this type of phase diagram are usually called 'monotectic alloys'.

further cooling the liquid L_1 decomposes at the monotectic temperature T_m into the solid phase A and the liquid L_2 of composition c_o. The alloy is now a mixture of solid A and liquid L_2 which will completely solidify, if it is cooled below the eutectic temperature (which often is equivalent to the melting temperature of B). The temperature difference between T_m and T_E can be quite large of the order of a few hundred degrees.

Looking at the energetics of immiscible alloys when a quench of a homogeneous liquid from a temperature above the binodal is performed to a temperature T_0 below, we have to differentiate between two possible situations. This is schematically shown in Fig.2 (calculated on the basis of a regular solution).

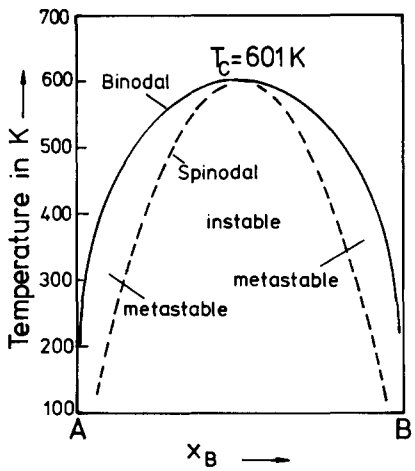

Fig.2 Miscibility gap and spinodal line in a regular solution system.

A shallow quench on the A-rich or B-rich side of the miscibility gap will make the liquid L metastable. The decomposition reaction $L \rightarrow L_1 + L_2$ proceeds then by the formation of nuclei which grow in a supersaturated matrix. The reaction is accompanied by the generation of heat. If we perform a deep quench or use alloys of near critical composition, then the liquid L is unstable against any concentration fluctuation and the decomposition proceeds without any nucleation barrier.The growth proceeds as an amplification of fluctuation waves, such that essentially one wavelength survives and a typical spinodal microstructure is obtained /3/.

Both situations will lead on earth to a rapid spatial separation of the liquids, since in metallic systems the liquids L_1 and L_2 normally exhibit appreciable density differences. In microgravity this can be prevented as shown with metallic immiscibles /2,5/ and with nometallics /3,6/ . We can basically devide the general sequence of forming a two phase alloy from a one phase one into three steps: 1. nucleation of the minority phase, 2. growth of the nuclei, 3. coarsening or Ostwald ripening. The driving force Δ G of transformation, which is the heat of reaction ΔH_{L1L2} multiplied by the undercooling

Δ T , ($\Delta G = \Delta H_{L_1L_2}\Delta T$), governs the nucleation process. It is partly consumed to overcome the nucleation barrier (formation of interfaces). Heaving a supersaturated matrix alloy after nucleation is completed growth of the nuclei will proceed in liquid immiscible metals by diffusion. The final step is then Ostwald ripening which dominates the process of coarsening, when the supersaturation has decreased to very low values. Then, the only driving force remaining is the high total interfacial energy of a dispersion when compared with one big droplet of the same volume .

Theories for all three steps are well developed in the literature and they should especially apply to describe the decomposition of liquid metals into two liquid phases.For example, these are the classical theory of nucleation by Becker and Vollmer, diffusional growth processes as calculated by Zener and coarsening by Ostwald ripening as described by the Lifshitz-Slyozov-Wagner theory (for a reference see /7/). In reality, there is no such clear distinction into three different steps. In the last ten years some unifying theories have been developed which calculate the evolution of the mean particle size and the particle size distribution from the very beginning of the decomposition process to its final stage linking all steps self-consistently together (Langer and Schwartz /8/.Kampmann and Wagner /9/).

In principle,all these theories assume material behaviour, which is perfectly established by liquids (isotropy of mass transport,of interfacial energy, highly mobile interfaces,to name a few), but they have overlooked one general feature of liquids compared to solids: generally, a liquid is not at rest in all its parts. On decomposition nuclei and droplets may move with respect to the matrix fluid (e.g. brownian motion); in the matrix fluid convective flows may occur, particles may settle (e.g. sedimentation in a gravity field) and relative motions between nuclei or droplets are possible leading to collision and coagulation events and thus additional coarsening of a dispersion. Today there is no theory available that takes into account all these special effects which occur in real liquids and which may determine the behaviour during the transition into a two phase alloy till solidification.

From the large number of experiments performed in space we will discuss four which cover the steps of decomposition mentioned above and show the action of convective motions occuring in liquids.

STUDIES ON NUCLEATION

The nucleation proces in liquid metallic alloys is very difficult to measure directly, in contrast to solid state reactions, where one can use for example electron microscopy, field ion microscopy with atom probe or small angle neutron scattering.This is in contrast to transparent materials, where optical techniques are quite informative (light scattering,photon correlation spectroscopy). The analysis of solidified specimens is very ambigious, because a solidification process redistributes solute and the droplets, even coagulation may occur changing the droplet size distribution and their locations. Therefore, indirect techniques are used, which have shown their value in liquid to solid transitions.

Phase decomposition in a liquid miscibility gap alloy was investigated by Perepezko,Galup and Cooper /10/. They studied Bi-Ga alloys of various compositions. The phase diagram is shown in Fig.3. The alloy has

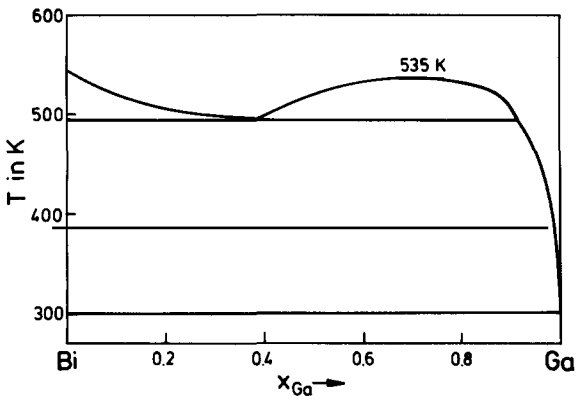

Fig.3 Phase diagram of Bi-Ga.

a flat binodal curve, with a critical point lower than the melting point of Bi and a very high concentration of Ga at the monotectic point. To observe the onset of nucleation they used liquid alloys heated to a temperature above the binodal line and then artficially emulsified them in an inert carrier liquid, with droplet sizes ranging from 5-10 μm. On cooling such a suspension of droplets being each a homogeneous liquid L each droplet nucleates the minority phase (L_1 or L_2). The temperature where the decomposition occurs is recorded by differential thermal analysis. The transisiton temperature depends to a certain degree on the radius of the droplets, in as much as the number of heteroge-

neous nucleation centres decreases with the droplet volume. Therefore, the smaller the droplets, the greater the possibility of achieving homogeneous nucleation. Their essential result is depicted in Fig.4, which shows the undercooling measured with respect to the binodal line (Fig.3) and its metastable extension on both sides, which could be measured and calculated (from an extended regular solution model).

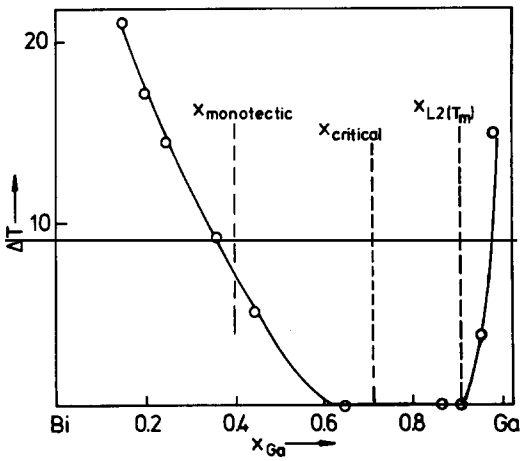

Fig.4 Measured undercoolings in Bi-Ga alloys. ΔT is the difference between the measured onset of decomposition and the extended miscibility gap.

The measured undercoolings are quite low compared to undercoolings observable in liquid-solid transitions with the same method (this may be explained by a low free energy barrier to nucleation). In addition, it is an interesting feature that the undercoolings achieved are quite different on both sides of the miscibility gap. Whereas the formation fo L_2 droplets from an L_1liquid requires some undercooling the formation of L_1 from L_2 liquids occurs without undercooling (at least as detectable by this method). This behaviour can be explained with the help of the phase diagram (Fig.3) and the phenomenon of critical wetting as proposed by Cahn /11/.

The phase diagram shows, that the miscibility gap is not symmetrical with respect to the critical concentration. This reflects the energetics of the system,i.e. the free enthalpy-concentration (G-x) curves.Predel and coworkers /12/ have shown by calorimetric measurements,that the Bi-Ga system shows deviations from a symmetrical regular solution. According to the theory of homogeneous nucleation, the gain in volume free energy on nucleation depends on the slope of the G-x-curves taken at the gross composition of the alloy. Therefore, the shape of the G-x-curves is important for the amount of free energy gain. Whenever the G-x-curves are asymmetric the undercooling necessary to induce nucleation should vary along the binodal line. The effect cannot be calculated for Bi-Ga, since the experimental data are not accurate enough.

The effect of critical wetting can be explained as follows. At the critical point of the miscibility gap the surface energy between the two liquids vanishes as well as the difference between the surface free energy between the liquids and a third phase like the droplet coating, which is here the inert carrier liquid. For two liquids in contact with the coating at a finite contact angle Θ the interfacial energies are related by Young's equation. If the interfacial energy of the two liquids σ_{L1L2} is lower than the difference between the interfacial energies of the two liquids and the coating material, then one liquid completely wets the coating. The other liquid is incorporated into it, such that no contact with the coating is possible. Cahn /11/ has pointed out, that the difference of the interfacial tensions of the liquids with the coating goes less rapidly to zero than the interfacial tension between the two liquids. Thus there should be a regime of critical wetting below the critical temperature of the miscibility gap. If this occurs on the Ga-rich side of the miscibility gap on reaching the binodal line L_1 wets completely the coating,thus L_1 is nucleated heterogeneously at the coating surface in a perfect way, i.e. without undercooling.

This example of an experimental study shows, that nucleation in liquid immiscibles shows new phenomena and some problems not met within liquid to solid transitions or solid state decompositions. Since there are no other investigation known on nucleation of a liquid metal in a liquid it is not clear, if this result can be generalized to all immiscibles, especially, since the characteristic features of the phase diagrams show considerable differences. The difficult experimental situation met with metallic immiscibles leaves some questions unanswered concerning nucleation: the nucleation rate, size distribution, the height of the nucleation barrier etc.. Probably investigations with transparent materials will improve our understanding, especially if they are chosen to correspond in some important aspects to metals (like succinonitril has improved our understanding of solidification processes).

STUDIES ON GROWTH

The experimental situation for the study of growth processes of nucleated dispersoids in metallic immiscibles is as difficult as mentioned above for the case of nucleation. There is, however, one experiment, in which droplet growth and dissolution has been observed with high accuracy. Otto /13/ studied the precipitation of liquid mercury from a supersaturated gallium melt in an automated MAUS facility on board the Space Shuttle.He used time temperature profiles such that the temperature cycled around the binodal in a controlled fashion. Therefore, both growth and dissolution took place. The dispersion of Hg droplets in Ga could be recorded in situ by X-ray transmission photographs. Stationary mercury droplets were observed with diameters in the range of 0.2 to 1.2 mm. Fig.5 shows such X-ray photographs of a cooling cycle, showing clearly

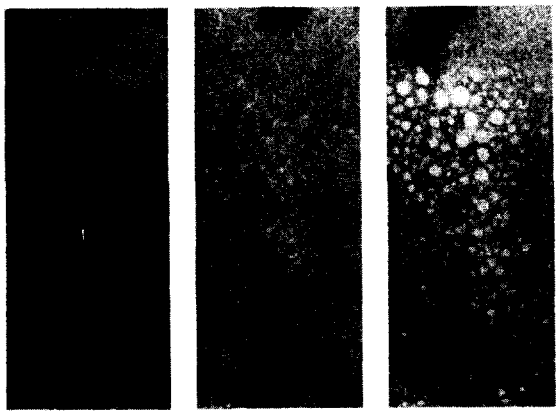

Fig.5 Precipitation process of mercury droplets from a supersaturated Ga(Hg)-liquid on cooling into the miscibility gap.

the appearence and growth of the droplets. During heating the same droplets disappear. X-ray photographs of the latter part of the experiment was analyzed with image analyzing techniques (grey density distributions around the droplets),in order to obtain information on the concentration field of mercury in the vicinity of the droplets. In addition a model was used to test theories of diffusional dissolution, expected to take place under microgravity conditions /14/.From time series of photographs of single droplets the time dependent decrease of droplet radius was evaluated. In the numerical analysis of the dissolution process Frohberg and Otto /14/ solved Fick's second law with appropriate boundary conditions, including the concentration changes due to the temperature change in the heating phase as given by the phase diagram. The temperature dependence of the diffusion coefficient was used according to the very careful measurements of Frohberg on tin and tin-indium alloys performed in Spacelab 1 and the D1-mission. The result of these calculations is shown

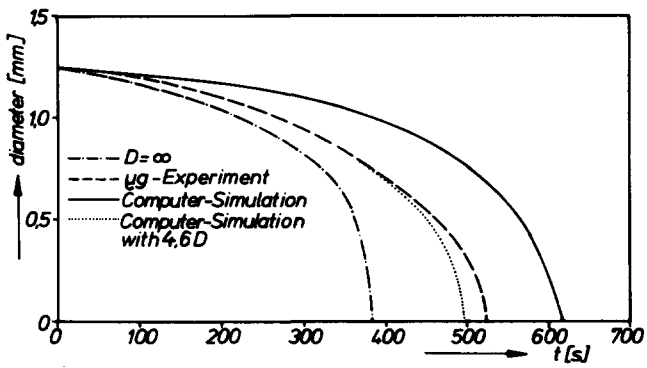

Fig.6 Time dependence of the radius of a dissolving Hg droplet in liquid Ga

in Fig.6 together with the experimental data. Obviously,the experimental values are closer to the quasi-equilibrium dissolution with $D = \infty$,than to the computer calculated curves using the finite most reliable diffusion coefficient. If one assumes a higher diffusion coefficient than calculated from the Frohberg data a good agreement between experiment and theory can be obtained.

This is in direct contrast of the expected bahaviour, that the effective diffusion coefficient should be smaller under microgravity conditions than under 1 g. This discrepancy has not been resolved up to now, but interfacial instabilities are discussed which may lead to small scale fluid motions and thus a convective contribution.

STUDIES ON COARSENING

The process of Ostwald ripening was studied by Kneissl and Fischmeister /15/ in a Spacelab 1 experiment and analyzed later by Ratke,Fischmeister and Kneissl /16/. They used Zn-Pb alloys with various Pb concentrations (2 - 5 wt% Pb). Alloys with a very fine microstructure of the dispersion were produced on earth by rapid quenching of homogeneous alloys ($\dot{T} > 100K/s$) and then heated in Spacelab 1 right into the miscibility gap to avoid nucleation and held isothermally (1h 475 ^{0} C) followed by a free cooling till solidification. Metallographic sections of the samples prior and post flight were analyzed in a SEM connected on-line to an automatic image analyzing system such that the section radii and their distribution could be evaluated with a large number of particles. Fig.7 shows a comparison of the dispersions prior and post flight.During the experiment the droplets have coarsened considerably (note the magnification).

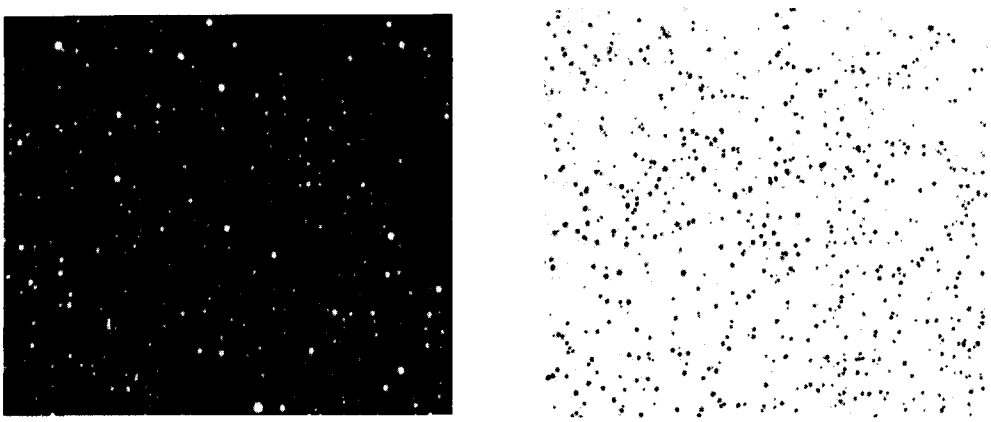

Fig.7 Metallographic sections of a Zn+4 wt.% Pb alloy prior (left) and post flight in Spacelab 1 (right). The magnification of the left picture is 600 X, that of the right 33 X.

In addition,it must be stated that all samples showed no massive segregation of the lead droplets, but the they were really homogeneously distributed throughout the samples (besides small scale inhomogeneities /16/). It was expected that after reaching the temperature of isothermal anneal the dispersion would coarsen by Ostwald ripening, because the concentration of solute atoms (Pb) should be determined only by the radius dependence of the chemical potential, giving rise to diffusion currents between small (high concentration of Pb) and big droplets (low concentration of Pb). Therefore the bigger ones should grow on the expense of the smaller ones. Since only one experimental run with one time for growth was available, a comparison between experiment and theoretical predictions made for this kind of coarsening by Lifshitz, Slyozov /17/ and Wagner /18/ can only be achieved via the particle distribution functions. Fig.8 shows the experimental particle distribution together with the theoretical one (LSW theory for diffusional mass transport) for alloys with 3 and 4 wt.% Pb (0.65 and .8 Vol.% Pb at the temperature of isothermal anneal). The comparison shows, that there are similiarities between both,especially at smaller particle diameters. There are, however, discrepancies at larger diameters and at the maximum. Possible disturbing effects yielding distributions differing from the LSW prediction are collision and coagulation events, shown in Fig.9.
There is one major source leading to droplet motion and therefore gives the possibility that coagulation may have taken place. During the solidification process there is a temperature gradient inside each sample leading to Marangoni motion of the droplets with different velocities because of their size spectrum. The cooling period in the Spacelab 1 experiment lasted approximately 200 seconds till the onset of solidification by reaching the monotectic temperature.The coagulation was calculated by solving the Smoluchowski-Müller equation, which describes this phenomenon /19/, with a collision cross section as determined by the Marangoni motion using best estimates of the material parameters and assuming that initially, i.e. at the end of the isothermal anneal,a LSW-distribution had been established in the liquid samples.The result of the LSW distribution perturbed in this way is also shown in Fig.8. Although the distributions changed by this effect come closer to the experimental ones, there are still discrepancies, especially at larger particle diameters.

The differences between experiment and theory can be reduced, if one assumes that on cooling the samples a shearing motion is established within them (with a maximum velocity of a few ten microns/second) leading to so-called gradient collisions between the particles. Performing similar

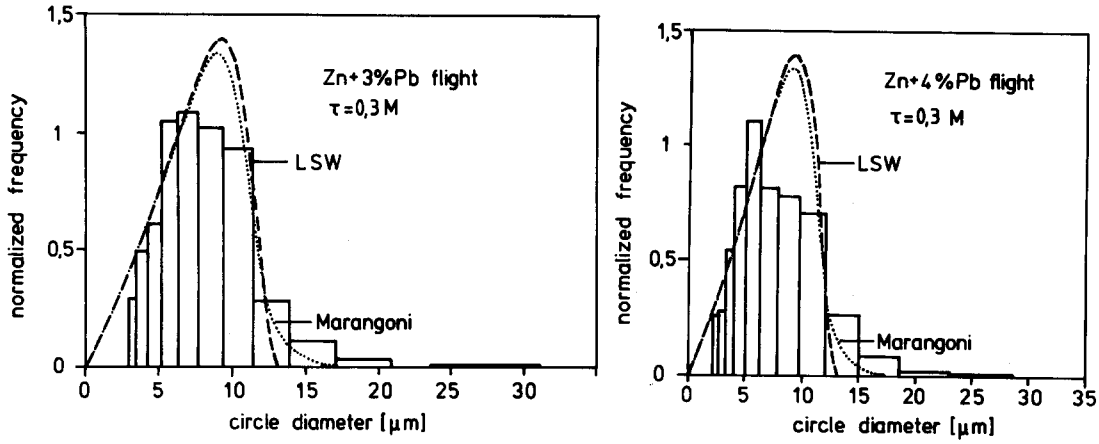

Fig.8 Distribution of lead droplet radii as determined on the metallographic sections (bars).Theoretical prediction by the LSW theory (- - -) converted from a 3-d radii distribution to a 2-d distribution of section radii by standard techniques of quantitative metallography. LSW-distribution changed by coagulation events due to Marangoni motion of the droplets (dotted curves).

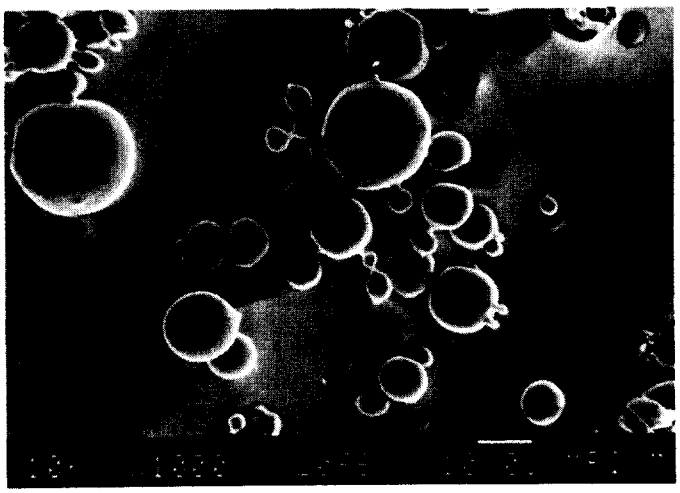

Fig.9 Coagulation events in an In-Al alloy rapidly quenched from the homogeneous state through the miscibility gap. The solidification front moves so fast, that it freezes the coagulation events.

calculations as for the Marangoni case gives results as shown in Fig.10.The agreement between theory and experiment is now considerably better, but this may be fortituous, since the origin of such a shearing motion of the matrix is unknown and therefore no reliable parameters are available to perform a calculation. The one shown here may be considered more as a fit than a real calculation.
From this experiment and its analysis the conclusions may be drawn, that coarsening in liquid immiscibles at the final stage of a precipitation process is probably well described by diffusional Ostwald ripening as predicted by the LSW-theory. There are,however, disturbing effects which stem from the need to solidify the sample in order to investigate their microstructure. This disadvantage can be converted into an advantage, since it offers the possibility to study coagulation processes which are always a necessary prerequisite for massive spatial separation of the phases. Future experiments planned for the EURECA-1 mission and an automated MAUS experiment with Al-Pb and Zn-Pb alloys will complete the Spacelab 1 measurements and extend them, since different experiment times are available improving the capabilities to compare theoretical predictions with the experimental findings.

STUDIES ON SEDIMENTATION

Sedimentation of droplets in a liquid dispersion - on earth caused mainly by gravity - is most often induced by temperature gradients leading to Marangoni motion of droplets from the cold to the hot side of a sample. This sedimentation was thought to be the major reason for the failures of many experiments performed with metallic immiscibles under microgravity during the last 15 years. For a discussion of this effect we choose a new experiment performed by Togano and coworkers /20/ in the

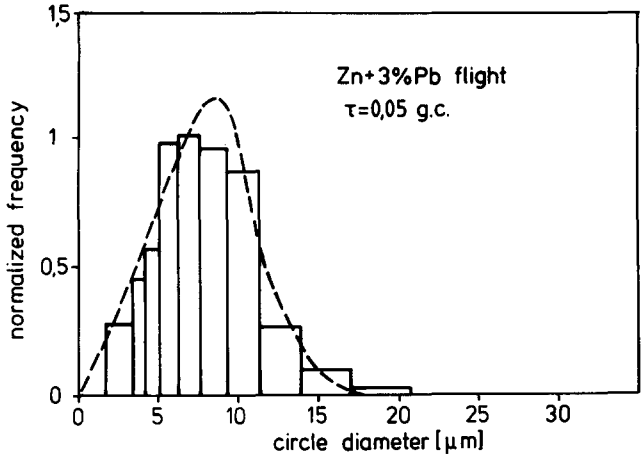

Fig.10 Change of an LSW distribution by gradient collisions due to a shearing motion inside the sample. All important physical parameters enter into the dimensionless time depicted in the figure. It serves here as a fit parameter.

TEXUS 13 mission with an Al+1 at.% Pb+1 at% Bi-alloy. They heated the alloy under microgravity conditions to a temperature above the binodal (homogenization) and cooled it till solidification with a huge temeprature gradient (up to 200 K/cm). They observed a strong sedimentation of the Pb-Bi droplets towards the hot side of the sample. Their result is shown in Fig.11 together with the relevant portion of the time-temperature profile. The volume content - equivalent to area fraction - of the Pb-Bi droplets was measured on a metallographic section.

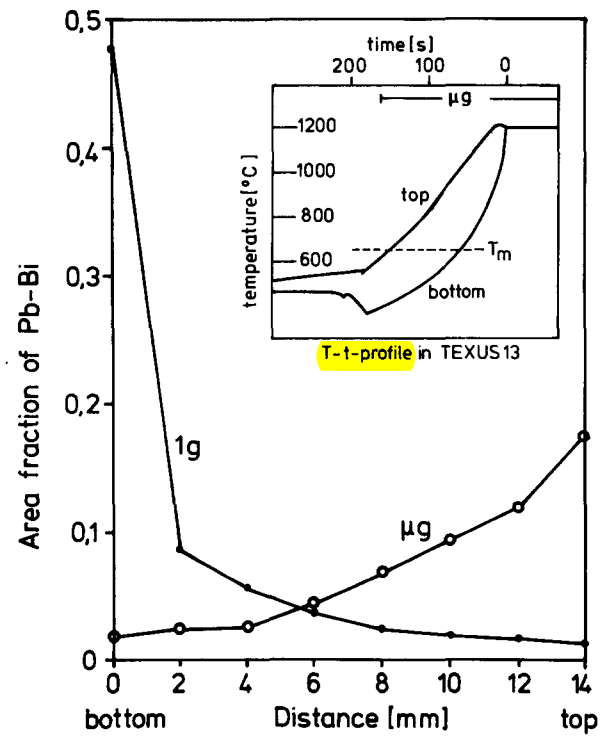

Fig.11 Volume fraction of dispersed Pb-Bi droplets in a Al matrix as a function of position in the sample as determined by Togano and coworkers /20/ under 1g and microgravity conditions of a TEXUS experiment. The insert shows a portion of the time temperature profile.

We have used this data to make a mathematical analysis. It was performed by solving the following continuity equation describing the concurrent growth of droplets due to some growth process and sedimentation by a position independent droplet velocity:

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial r}(\dot{r} f) + \frac{\partial}{\partial x}(u_{Marangoni} f) = 0$$

Diffusional growth of the droplets was assumed and described by the well known Zener parabolic growth law /7/. The Marangoni velocity was calulated according to the Young,Goldstein and Block equation (YGB) /21/. Nucleation was assumed to have occured heterogeneously at nucleation centers distributed in the sample according to a gaussian distribution. The initial particle radii distribution was assumed to be log-normal with a mean radius of 10 nm. Diffusion constant, supersaturation and the material parametres that enter the YGB equation were taken from the literature (or estimated).The volume fraction in each section of the sample could then be calculated numerically. The result of such a computer analsis is shown in Fig.12. There is a good

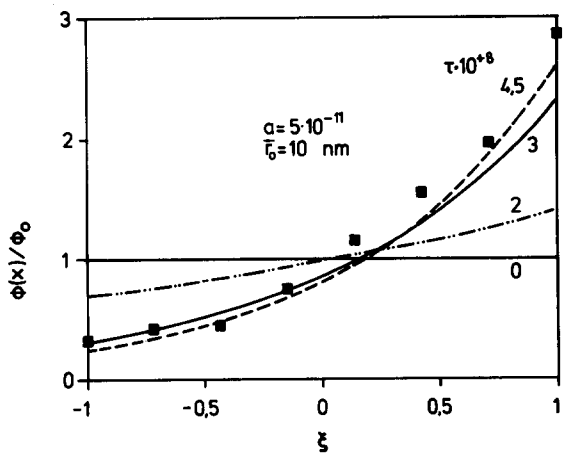

Fig.12 Numerical solution of concurrent diffusional growth and sedimentation due to Marangoni motion for the Al-Pb-Bi alloy used by Togano /20/. The squares represent the experimental results. The time is dimensionless and corresponds at its latest value to the experimental time. Initially the sample is nearly free of droplets, since they nucleate heterogeneously. The relative value of volume content is nevertheless equal to one initially.

agreement between calculation and experiment, such that we are convinced, that the inhomogeneuos distribution of droplets in the sample can really be attributed to Marangoni motion.

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